A SHEAF ON A COLIMIT OF SITES IS THE COLIMIT OF ITS PUSHFORWARDS AND PULLBACKS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Composition of counits of adjunctions. Categories, Lemma categories-lemma-compose-counits. Supplies the cocycle condition satisfied by the canonical maps.
- The identification of a sheaf with the colimit of its pullbacks along the transition morphisms. Same chapter, Lemma lemma-colimit. Supplies the description of the values of the sheaf as a filtered colimit, which is compared with the colimit in the statement.

Source reference. Original lemma 09YN: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2905–2905. Navigation pointer invoked at proof line 4225; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0DV0: sources/stacks-sites/pinned-original-source.tex, source lines 4242–4242. Navigation pointer invoked at proof line 4242; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Situation 0.1. Here we are given

(1) a cofiltered index category $\mathcal{I}$,

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(2) for $i \in \mathrm{Ob}(\mathcal{I})$ a site $\mathcal{C}_i$ such that every covering in $\mathcal{C}_i$ has a finite index set,
(3) for a morphism $a : i \to j$ in $\mathcal{I}$ a morphism of sites $f_a : \mathcal{C}_i \to \mathcal{C}_j$ given by a continuous functor $u_a : \mathcal{C}_j \to \mathcal{C}_i$,

such that $f_a \circ f_b = f_c$ whenever $c = a \circ b$ in $\mathcal{I}$.

Lemma 0.2. *In Situation 0.1 assume we have a sheaf $\mathcal{F}$ on $\mathcal{C}$. Then*

$$\mathcal{F} = \operatorname{colim} f_i^{-1} f_{i,*} \mathcal{F}$$

where the transition maps are $f_j^{-1}\varphi_a$ for $a : j \to i$ where $\varphi_a : f_a^{-1} f_{i,*}\mathcal{F} \to f_{j,*}\mathcal{F}$ is a canonical map satisfying a cocycle condition as in Lemma 09YN.

Proof. For the morphism

$$\varphi_a : f_a^{-1} f_{i,*}\mathcal{F} \to f_{j,*}\mathcal{F}$$

we choose the adjoint to the identity map

$$f_{i,*}\mathcal{F} \to f_{a,*} f_{j,*}\mathcal{F}$$

Hence φ_a is the counit for the adjunction given by $(f_a^{-1}, f_{a,*})$. We must prove that for all $a : j \to i$ and $b : k \to i$ with composition $c = a \circ b$ we have $\varphi_c = \varphi_b \circ f_b^{-1}\varphi_a$. This follows from Categories, Lemma 0DV0. Lastly, we must prove that the map given by adjunction

$$\operatorname{colim}_{i \in I} f_i^{-1} f_{i,*}\mathcal{F} \longrightarrow \mathcal{F}$$

is an isomorphism. For an object U of $\mathcal{C}$ we need to show the map

$$(\operatorname{colim}_{i \in I} f_i^{-1}\mathcal{F}_i)(U) \to \mathcal{F}(U)$$

is bijective. Choose an i and an object U_i of $\mathcal{C}_i$ with $u_i(U_i) = U$. Then the left hand side is equal to

$$(\operatorname{colim}_{i \in I} f_i^{-1}\mathcal{F}_i)(U) = \operatorname{colim}_{a : j \to i} f_{j,*}\mathcal{F}(u_a(U_i))$$

by Lemma 09YN. Since $u_j(u_a(U_i)) = U$ we have $f_{j,*}\mathcal{F}(u_a(U_i)) = \mathcal{F}(U)$ for all $a : j \to i$ by definition. Hence the value of the colimit is $\mathcal{F}(U)$ and the proof is complete. □